IMMFLY

N053AV (S/N: W14061878622) — Disk Critical Medium Error Diagnostic Report

Immfly Ground Operations / IT — Session: September 1, 2026

Field	Value
Machine	N053AV (S/N: W14061878622)
Affected component	Internal NVMe disk — active, recurring read failure at a fixed sector
Initial symptom	Repeated kernel-level critical medium errors observed during a routine login session
Affected location	Sector 167,843,888 (and a nearby sector, 167,843,872) — inside the filesystem, not the boot/partition-table sector
Unit status	Unit boots and is otherwise operational (network connectivity confirmed via DHCP), unlike sector-0 failures seen on other units
Error pattern	The same sector was repeatedly re-read and re-failed every few seconds throughout the session, indicating an active retry loop, not a one-time event
SMART data	Not collected — smartmontools was not installed on the unit and could not be installed via apt due to no internet access on this unit at the time
Recommended action	Preventive disk replacement recommended; if not replaced yet, capture full SMART data at the next opportunity (via the offline package bundle) to quantify severity and locate the affected file/data

Narrative Summary

During a routine login session on N053AV, the kernel log showed repeated `critical medium error` messages on the internal NVMe disk (`nvme0n1`), all pointing to sector 167,843,888 (with one adjacent occurrence at sector 167,843,872). Unlike the sector-0 failure previously documented on N148AV — which affects the boot/partition-table sector and renders the disk unusable for booting or imaging — this error is located well inside the filesystem, so the unit was able to boot normally and remained network-reachable throughout the session (a DHCP lease was successfully obtained on br0).

Notably, the identical error recurred repeatedly, every few seconds, throughout the session — including while the user was simply typing unrelated commands — indicating some process or service on the system is actively and repeatedly attempting to read the damaged sector, rather than this being a single historical event. This points to an active, ongoing read failure affecting whatever file or data structure lives at that location, which will likely continue to generate errors (and may cause application-level failures or data corruption for whatever content is stored there) until the disk is replaced.

A full SMART health report was not obtained during this session, as smartmontools was not already installed on the unit and could not be installed via the package manager due to no internet connectivity being available on N053AV at the time (DNS resolution to the Ubuntu package repositories failed). The fleet's offline dependency bundle (`modem_fix_offline_packages_focal.tar.gz`), which includes smartmontools as a bonus package, was identified as the fastest path to install it at the next opportunity without requiring internet access.

Conclusion and Recommendation

N053AV shows an active, recurring disk read failure at a fixed internal sector. While the unit remains bootable and network-reachable — a less severe symptom than the sector-0 failures seen elsewhere in the fleet — the repeated, ongoing nature of the error indicates real, active media damage rather than a transient glitch, consistent with the disk-degradation pattern previously documented on N070AV.

Recommended action: Preventive disk replacement is recommended given the confirmed active failure. If replacement is not immediately available, install smartmontools from the offline package bundle and capture a full SMART report to quantify the drive's overall health (media error count, unsafe shutdown count, percentage used) and confirm whether additional sectors are affected beyond the one already observed.

Appendix: Kernel Log Excerpt

```
blk_update_request: critical medium error, dev nvme0n1, sector 167843872 op 0x0:(READ) flags 0x80700 phys_seg 2
prio class 0
blk_update_request: critical medium error, dev nvme0n1, sector 167843888 op 0x0:(READ) flags 0x0 phys_seg 1
prio class 0
(repeated at least 10 times across a ~40-second window during a single, otherwise idle terminal session)
```